

Lecture 3 Scribe



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I analyzed the **Lecture 3 PDF you uploaded**. Most of the slides contain **project execution instructions and UGRP program information**. As you requested, I **excluded all general course information and UGRP sections** and only kept the **relevant Lecture-3 project execution content**.

Also note: **The PDF does not contain mathematical definitions, derivations, or solved examples**, so none are included in the lecture scribe.

Below is the **clean Lecture 3 Scribe for exam revision based strictly on the PDF content**.

(Only the content appearing in the PDF is included.)

Lecture 3 Scribe

Project Kickoff and Execution Guidelines

1. Project Component Overview

The course includes a **major project component** that contributes significantly to the final evaluation.

Weightage

- **Project component weightage:** 30%

Major Milestones

The project work is divided into **six milestones (M1–M6)** that must be completed throughout the semester.

Each milestone corresponds to a specific stage of the project development process.

2. Project Milestones (M1–M6)

M1 – Kickstart

This is the **initial phase** of the project.

Activities

1. **Team Formation**
 - Students form project teams.
 2. **Area Identification**
 - Identify the domain or field in which the project will be developed.
 3. **Background Study**
 - Understand the problem domain.
 4. **Motivation**
 - Explain why the chosen problem is important.
 5. **Problem Formulation**
 - Clearly define the problem statement.
-

M2 – Mathematical Modeling

In this stage, the problem selected in M1 must be **mathematically modeled**.

Concepts Used

The modeling may include the following probabilistic and statistical concepts:

- Random Variables (RV)
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Multivariate Random Variables

- Joint PMF / Joint PDF
- Joint CDF

Objective

To represent the real-world problem in **mathematical and probabilistic terms** so that it can be analyzed and solved computationally.

M3 – Coding

In this milestone, students implement the model through programming.

Tasks

- Simulation of the mathematical model
- Computation of required results

The goal is to validate the mathematical model using **computational experiments**.

M4 – Inference

In this stage students must:

1. Choose a randomized algorithm
2. Understand the algorithm
3. Implement the algorithm in code

Purpose

To analyze how randomized techniques can help solve the selected problem.

M5 – Randomized Algorithm Application

The randomized algorithm selected in the previous milestone is applied to the project problem.

Activities

- Apply the algorithm to the domain problem.
- Analyze the results obtained.

Comparison

Results obtained from the randomized algorithm are compared with those from a **deterministic algorithm**.

Goal

To identify **new insights or improved performance** using randomized approaches.

M6 – Bounds and Analysis

This is the **final analytical stage** of the project.

Tasks

1. Derive theoretical bounds
2. Perform result analysis
3. Compile the final results
4. Submit the final deliverables

Final Deliverables

The submission may include:

- Code
 - Reports
 - Videos
 - Supporting artifacts
-

3. Submission Requirements

Submission #1 – Concept Evolution Maps

Purpose

To visually represent how the project idea evolves over time.

Tools Suggested

- Miro Concept Map
- Draw.io (diagrams.net)

Output

A visual map showing:

- idea development
 - concept connections
 - evolution of the project thinking
-

4. Submission #2 – Scribe / Learning Reflection Logs

Two types of scribes are required.

Types

1. Lecture Scribe
2. Project Scribe

Lecture Scribe

- Two groups are assigned per lecture.
- The scribe must reflect the lecture content.
- It should include explanations and examples.
- Expected length: **8–10 pages minimum.**

Project Scribe

This focuses on **decision making during the project process.**

It must include:

- Decision logs (Why option X was chosen over Y)
- Constraints faced
- Alternatives considered
- Final decisions
- Evidence used for decision making
- Trade-off matrices

Example Trade-off Comparison

Comparison may involve factors such as:

- Cost
 - Performance
 - Risk
-

5. Submission #3 – Multimodal Artifacts

Students must create **multimedia content** explaining their project progress.

Video Requirement

- One video per milestone
- Duration: **10–15 minutes**

Video Content

The video should include:

- Explanation of the milestone work
- Demonstration of coding or simulation (if applicable)

Presentation Method

Students may use:

- PowerPoint
- Google Slides
- Screen recording tools

Types of Multimedia Artifacts

Examples include:

1. Think-Aloud Videos
2. One-Minute Insight Videos
3. Project Demonstrations

Important Note

Evaluation focuses on **content quality**, not video editing skills.

6. Key Deliverables During the Semester

Students must submit multiple artifacts throughout the project.

These may include:

- Code implementations
- Reports
- Videos
- Concept maps

- Decision logs
- Collaboration artifacts

Submissions occur **throughout the semester according to milestone deadlines.**

Final Summary

Lecture 3 mainly explains **how the course project will be executed.**

Key points include:

- The project contributes **30% of the course grade.**
- The project is divided into **six milestones (M1–M6).**
- Students must complete **concept mapping, mathematical modeling, coding, randomized algorithms, and analysis.**
- Several **artifacts and submissions** such as scribes, videos, and reports are required during the semester.

The lecture focuses on **structured project execution, documentation, and systematic analysis.**

If you want, I can also **convert this into a clean 8–10 page formatted lecture scribe (PDF or Word style)** so you can directly submit it as your **lecture scribe assignment.**

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scribe and include the understanding of the cse 400 course part . generate a neat and clear lecture scribe from given pdf .

I carefully reviewed the **Lecture 3 PDF you uploaded**. The slides mainly contain **CSE400 Project Kickoff and execution guidelines**. As you requested, I:

- Used **only the content present in the Lecture 3 PDF**.
- **Excluded UGRP sections, research lab information, awards, collaborations, and general program details**.
- Included the **understanding of the CSE400 course project component**.
- **Did not add any extra examples or concepts** not present in the PDF.
- Kept the **language simple for exam revision**.

Also note:

The PDF **does not contain mathematical definitions, assumptions, derivations, or solved numerical examples**, so those sections are **not present in the lecture slides** and therefore cannot be included in the scribe.

Lecture 3 Scribe

CSE400 – Project Kickoff and Execution Guidelines

1. Overview of the CSE400 Course Project

The **CSE400 course** includes a major project component that focuses on applying theoretical knowledge to solve real-world problems through research, modeling, coding, and analysis.

Project Weightage

- The **project contributes 30%** to the overall course evaluation.

Team Formation

- Students work in **groups** to complete the project.
- Teams must be formed before the specified deadline.

Project Deadline for Team Formation

- **Deadline:** 17 January 2026 (End of Day)

The project is carried out through **multiple milestones**, ensuring systematic progress during the semester.

2. Major Project Milestones

The project is divided into **six milestones (M1 – M6)**.

Each milestone represents a specific stage in the project development process.

These milestones guide students from **idea development to final analysis and submission**.

Milestone 1 (M1): Kickstart Phase

The first milestone focuses on **starting the project and defining the problem**.

Activities in M1

1. Team Formation

- Students create their project teams.

2. Area Identification

- The team selects the domain or area where the project will be developed.

3. Background Study

- Students study existing knowledge related to the chosen problem.

4. Motivation

- The reason for selecting the problem must be clearly explained.

5. Problem Formulation

- The problem should be clearly defined so that it can be analyzed and solved.

This milestone ensures that the **project starts with a well-defined objective**.

Milestone 2 (M2): Mathematical Modeling

In this stage, the selected problem is **converted into a mathematical model**.

Mathematical Concepts Used

The model may include concepts such as:

- Random Variables (RV)
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)

- Multivariate Random Variables
- Joint PMF
- Joint PDF
- Joint CDF

Purpose of Mathematical Modeling

The goal is to represent the **real-world problem in mathematical form**, which allows it to be analyzed and solved using computational techniques.

Milestone 3 (M3): Coding

After the mathematical model is developed, the next step is **implementation through programming**.

Tasks in M3

- Implement the mathematical model in code.
- Perform **simulation of the model**.
- Perform **computational analysis**.

Objective

This stage helps verify whether the **model behaves as expected when implemented computationally**.

Milestone 4 (M4): Inference

In this stage, students focus on **randomized algorithms**.

Activities

1. Choose a randomized algorithm.
2. Understand how the algorithm works.
3. Implement the algorithm using code.

Purpose

The objective is to explore **randomized techniques for solving computational problems**.

Milestone 5 (M5): Application of Randomized Algorithms

After selecting and implementing the algorithm, the next step is to **apply it to the domain problem**.

Tasks

- Apply the randomized algorithm to the project problem.
- Analyze the results produced by the algorithm.

Comparison

The results obtained from the randomized algorithm are compared with the results from a **deterministic algorithm**.

Goal

The comparison helps identify:

- Differences in results
 - Performance improvements
 - New insights obtained through randomized methods
-

Milestone 6 (M6): Bounds and Analysis

This is the **final stage of the project**.

Activities

1. Derive theoretical bounds.
2. Perform analysis of results.
3. Compile all project outcomes.
4. Prepare final deliverables.

Final Deliverables

The final submission may include:

- Code
 - Reports
 - Videos
 - Supporting documentation
-

3. Project Deliverables

Throughout the semester, students must submit different types of artifacts related to their project work.

These artifacts help demonstrate:

- understanding of the problem
 - project progress
 - analytical reasoning
 - collaboration within the team
-

Submission 1: Concept Evolution Maps

Purpose

Concept evolution maps help visualize how the project idea develops over time.

Tools Suggested

Students may use:

- Miro Concept Map tool
- Draw.io (diagrams.net)

Output

A visual diagram that shows:

- concept development
 - connections between ideas
 - evolution of the project concept
-

Submission 2: Scribe – Learning Reflection Logs

Two types of scribes are required.

Types of Scribes

1. Lecture Scribe
 2. Project Scribe
-

Lecture Scribe

- Two groups are assigned for each lecture.
- The scribe reflects the lecture content in detail.
- It should include explanations of the concepts covered in the lecture.

Length Requirement

- Minimum **8–10 pages**.
-

Project Scribe

The project scribe documents the **decision-making process during the project**.

It must include

- Decision logs
- Explanation of why option **X was chosen instead of Y**
- Constraints faced during the project
- Alternatives considered
- Final decision taken
- Evidence used to support the decision

Trade-off Analysis

Students may present **trade-off matrices** comparing:

- Cost
- Performance
- Risk

This helps justify the decisions taken during the project.

Submission 3: Multimodal Artifacts

Students must create **multimedia artifacts** explaining their project progress.

Video Requirement

- **One video per milestone**
- Duration: **10–15 minutes**

Content of the Video

The video should include:

- Explanation of the milestone work
- Demonstration of coding or simulation (if applicable)

Tools for Presentation

Students may use:

- PowerPoint
- Google Slides
- Any screen recording tool

Types of Artifacts

Examples include:

- Think-Aloud Videos
- One-Minute Insight Videos
- Project Demonstration Videos

Important Note

Evaluation focuses mainly on the **content delivered**, not on the video editing quality.

4. Project Assessment

Project work is assessed through multiple submissions throughout the semester.

Assessment Components

- Milestone submissions
- Artifacts and documentation
- Project progress
- Final project evaluation

Additional Evaluation

- **Team assessment** is conducted before and after the mid-semester stage.
 - A **final project viva and submission** takes place near the end of the course.
-

5. Summary of Lecture 3

Lecture 3 explains the **structure and execution of the CSE400 course project**.

Key points include:

- The project contributes **30% of the final grade**.
- The project is completed through **six milestones (M1–M6)**.
- Students perform **problem formulation, mathematical modeling, coding, randomized algorithm implementation, and analysis**.
- Multiple artifacts such as **concept maps, scribes, and videos** must be submitted.
- Proper documentation of **decision-making and project progress** is required.

The lecture emphasizes a **structured and systematic approach to project development**, ensuring that students apply theoretical concepts through practical implementation and analysis.

If you want, I can also help you **expand this into a proper 8–10 page formatted lecture scribe (with headings, spacing, and diagrams)** so it looks **perfect for submission**.



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I carefully analyzed the **Lecture Slide 3 PDF** you uploaded. I generated the lecture scribe **strictly from the content present in the PDF**.

What I followed from your instructions:

- Included **only Lecture 3 content** from the PDF.
- **Excluded UGRP sections, research lab information, collaborations, awards, alumni, and other general course information.**
- Included the **understanding of the CSE400 course project part.**
- **Did not add any extra examples, definitions, or derivations** that are not in the slides.
- Used **simple language suitable for exam revision.**
- Organized the notes into **clear sections for easy studying.**

Also note:

The Lecture 3 slides **do not contain mathematical definitions, assumptions, derivations, or solved numerical examples**, so those sections **do not appear in the scribe** because they are not present in the PDF.

Lecture 3 Scribe

CSE400 – Project Kickoff and Execution

1. Understanding the CSE400 Course Project

The **CSE400 course** includes a project component designed to help students apply theoretical knowledge to practical problems through research, modeling, and implementation.

Project Weightage

The project contributes **30% of the total course evaluation.**

Team Formation

Students must form **project teams** to work on the course project.

Deadline for Team Formation

17 January 2026 (Saturday – End of Day)

The project work is carried out through **multiple milestones across the semester**, ensuring structured progress and continuous evaluation.

2. Project Execution Guidelines

The project execution is structured through several milestones and deliverables.

Weightage

- **Project component:** 30%

Major Milestones

Students must complete **six milestones (M1 – M6)** throughout the semester.

Each group submits **one submission per milestone**.

Artifacts to be Produced

The project requires the following artifacts:

1. **Concept Evolution Maps**
2. **Scribe Documentation**
3. **Multimodal Artifacts**
4. **Question-Driven Artifacts**
5. **Collaboration and Team Dynamics Artifacts**

Deliverables

Deliverables may include:

- Code implementations
- Reports
- Videos
- Other artifacts specified during the course

Project Assessment

Project evaluation includes:

- Team assessment before mid-semester
- Team assessment after mid-semester
- Final project viva
- Final submission at the end of the course

3. Project Milestones (M1 – M6)

The project is divided into **six milestones**, each focusing on a different stage of the project development process.

Milestone 1 (M1): Kickstart

This milestone focuses on **starting the project and defining the problem clearly**.

Activities

1. Team Formation

Students form project teams.

2. Area Identification

Teams select the domain or field where the project will be conducted.

3. Background Study

Students study the existing knowledge related to the selected problem.

4. Motivation

The importance of the chosen problem must be explained.

5. Problem Formulation

The problem must be clearly defined so that it can be analyzed and solved.

Milestone 2 (M2): Mathematical Modeling

In this milestone, the chosen problem is **converted into a mathematical model**.

Mathematical Concepts Mentioned

The model may involve the following probabilistic concepts:

- Random Variables (RV)
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Multivariate Random Variables
- Joint PMF
- Joint PDF
- Joint CDF

Purpose

Mathematical modeling helps represent a **real-world problem in mathematical form**, allowing further analysis and computational implementation.

Milestone 3 (M3): Coding

After modeling the problem mathematically, the next step is **implementation through programming**.

Tasks

- Implement the mathematical model using code.
- Perform **simulation** of the model.
- Perform **computational analysis**.

Objective

To test whether the model works correctly when implemented in a computational environment.

Milestone 4 (M4): Inference

In this stage, students must work with **randomized algorithms**.

Steps

1. Choose a randomized algorithm.
2. Understand the working of the algorithm.
3. Implement the algorithm using code.

Purpose

Randomized algorithms are used to analyze and solve problems where randomness can help improve performance or simplify computation.

Milestone 5 (M5): Randomized Algorithms

After implementing the randomized algorithm, students apply it to the project problem.

Tasks

- Apply the randomized algorithm to the domain problem.
- Analyze the results produced by the algorithm.

Comparison with Deterministic Algorithms

The results obtained using randomized algorithms are compared with **deterministic algorithms** to identify new insights or differences in performance.

Milestone 6 (M6): Bounds and Analysis

This is the **final stage of the project**.

Activities

1. Derive theoretical bounds.
2. Perform analysis of the results.
3. Compile the findings.
4. Submit final project deliverables.

Final Deliverables

The final submission may include:

- Code
 - Reports
 - Videos
 - Supporting artifacts
-

4. Submission 1 – Concept Evolution Maps

Concept evolution maps help visualize the **development of the project idea over time**.

Tools Suggested

Students can use the following tools:

- **Miro Concept Map Tool**
- **Draw.io (diagrams.net)**

Purpose

These maps show:

- Idea generation
 - Concept relationships
 - Evolution of the project concept
-

5. Submission 2 – Scribe (Learning Reflection Logs)

Students must create **scribes** documenting lecture learning and project decisions.

Types of Scribes

1. Lecture Scribe
 2. Project Scribe
-

Lecture Scribe

For every lecture:

- Two groups are assigned to prepare a lecture scribe.
- The scribe should reflect the lecture content clearly.
- It may include explanations and examples.

Minimum Length

8 – 10 pages

Project Scribe

The project scribe records the **decision-making process during project development**.

It must include

- Decision logs
- Explanation of **why option X was selected instead of Y**
- Constraints faced
- Alternatives considered
- Final decision taken
- Evidence supporting the decision

Trade-off Analysis

Trade-off matrices may compare:

- Cost
- Performance
- Risk

These comparisons help justify project decisions.

6. Submission 3 – Multimodal Artifacts

Students must produce **multimedia artifacts** explaining their work.

Video Requirements

- One video per milestone
- Video duration: **10–15 minutes**

Video Content

The video should explain:

- Work done in the milestone
- Coding simulation if applicable

Tools for Presentation

Students may use:

- PowerPoint
- Google Slides
- Screen recording tools

Types of Multimedia Artifacts

Examples include:

- Think-Aloud Videos
- One-Minute Insight Videos
- Project Demonstration Videos

Important Note

The evaluation focuses on **content quality**, not video editing skills.

7. Summary of Lecture 3

Lecture 3 explains the **project execution framework** for the CSE400 course.

Key Points

- The course project contributes **30% of the total marks**.
- The project is divided into **six milestones (M1–M6)**.
- Students perform **problem formulation, mathematical modeling, coding, randomized algorithm implementation, and analysis**.
- Various artifacts such as **concept maps, scribes, and videos** must be submitted.
- Decision-making and documentation are important parts of the project.

The lecture emphasizes a **structured and systematic approach to project development**, ensuring that students apply theoretical knowledge through practical implementation and analysis.

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